



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECI1143 SECTION 2

PROBABILITY AND STATISTICAL DATA ANALYSIS

PROJECT 2

NAME	MATRIC ID
AHMAD ZIYAAD BIN MOHD ABBAS	A23CS0206
AFIF SHAQIR IRFAN BIN ARQAM	A23CS0204
IMAN ABADI BIN MOHD NIZWAN	A23CS0084
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	A23CS0250

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1.0 INTRODUCTION

Globally, cardiovascular diseases (CVDs) account for 17.9 million deaths annually, making them the leading cause of death. Heart and blood vessel illnesses together referred to as CVDs include rheumatic heart disease, coronary heart disease, and cerebrovascular disease. This significant number raises the question as to why a large portion of people died from CVDs. Specifically, heart attacks, or myocardial infarctions are looked into for greater understanding of their causes and possible protective factors. We anticipated that the data would help us identify important heart attack risk factors. Age, blood pressure, cholesterol, smoking status, and family history were among the variables we expected to show up as strong signs.

2.0 DATASET

2.1 Description of Data Used

Population: Medical records from patients around the globe with the presence or absence of a heart attack risk

Sample: 999 patients

Data Description :

Variables	Type of Variable	Measurement Level
Cholesterol - Cholesterol levels of the patient	Quantitative	Ratio
Obesity - Obesity status of the patient (1: Obese, 0: Not obese)	Qualitative	Ordinal
Exercise Hour Per Week - Number of exercise hours per week	Quantitative	Ratio
BMI - Body Mass Index (BMI) of the patient	Quantitative	Ratio
Sex - Sex of the patient	Qualitative	Nominal
Heart rate - Heart rate of the patient	Quantitative	Ratio

2.2 Statistical Test Analysis

Selected Variables	Objectives	Test Analysis and Expected Outcome
Cholesterol, Obesity	To determine if the cholesterol levels of obese patients and non-obese patients are different	Analysis : 2 Sample Hypothesis Testing (Test on Mean, Variance Unknown) Expected Outcome : By using two sample t-tests, mean cholesterol levels of obese patients and non-obese patients can be compared. If the p-value is significant ($p < 0.05$), it suggests that there is significant difference in cholesterol level between the two groups.
Exercise Hour Per Week,BMI	Measures the strength of the association (linear relationship) between two variables.	Analysis: Correlation analysis Expected Outcome: By calculating the correlation coefficient using Pearson product-moment correlation coefficient between exercise hour per week and BMI.From this correlation analysis,we can determine the strength and direction of the relationship.
Cholesterol, Heart rate	To determine if the Cholesterol level affect the heart rate value of the patient	Analysis: Regression analysis Expected outcome: By performing a simple linear regression analysis, we can determine the relationship between heart rate and cholesterol If the regression coefficient is 5 statistically significant (typically $p < 0.05$),it would suggest that at heart rate we can quantify the extent of this relationship through this regression equation.

3.0 DATA ANALYSIS

3.1 Two Sample Hypothesis Test

In this analysis, variables Cholesterol and Obesity are used to test whether the cholesterol levels of obese patients and non-obese patients are different at a 95% confidence level, assuming unequal variances. From the data, number of samples (n), mean ($\bar{x}$), and standard deviation (s) can be calculated.

```
Descriptive statistics by group
group: 0
  vars   n   mean    sd median trimmed   mad min max range skew kurtosis   se
x1     1 523 260.45 80.31   258  260.62 96.37 120 400   280 0.01   -1.14 3.51
-----
group: 1
  vars   n   mean    sd median trimmed   mad min max range skew kurtosis   se
x1     1 476 253.5 80.89   247  252.74 98.59 120 400   280 0.1   -1.15 3.71
```

Categories	Non-Obese (0)	Obese(1)
Number of Samples	$n_1 = 523$	$n_1 = 476$
Mean	$\bar{x}_1 = 260.45$	$\bar{x}_1 = 253.5$
Standard Deviation	$s_1 = 80.31$	$s_2 = 80.89$

1. Hypothesis Statement :

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

μ_1 = mean of cholesterol levels of non-obese patients

μ_2 = mean of cholesterol levels of obese patients

2. Test Statistics :

With the confidence level of 95%, $\alpha = 0.05$. The test statistics, t_0 can be calculated using the formula :

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

By using RStudio, the results of the calculations are as follows,

```
welch Two Sample t-test

data: data2$cholesterol by data2$obesity
t = 1.3608, df = 986.82, p-value = 0.1739
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -3.072032 16.969938
sample estimates:
mean in group 0 mean in group 1
    260.4532      253.5042
```

The test statistics, $t_0 = 1.3608$ and the degree of freedom, $df = 986.2$. The probability, $p\text{-value} = 0.1739$.

3. Conclusion :

Since the $p\text{-value} > \alpha$, ($0.1739 > 0.05$), the null hypothesis, H_0 is failed to be rejected.

There is not enough evidence to conclude that there are differences between the mean of cholesterol levels of non-obese patients and mean of cholesterol levels of obese patients. Supposedly, obese people are more likely to have higher cholesterol levels. However there are other key factors that contribute to higher cholesterol levels such as smoking, drinking alcohol and the lack of exercises.

3.2 Correlation Test

In this part, we are using the variables to determine whether there is a linear relationship between exercise hours per week and BMI at a 95% confidence level using Pearson's Product-Moment Correlation Coefficient.

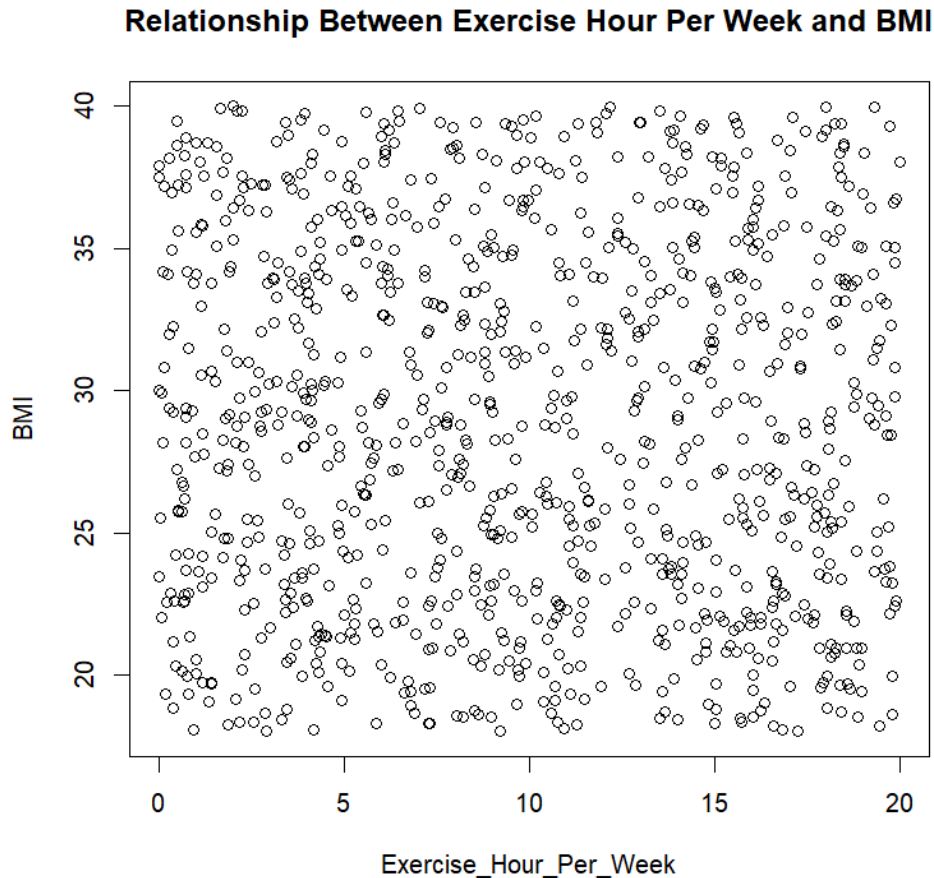


Figure 3.2 Correlation Analysis using R Programming

Figure 3.2 is "Relationship Between Exercise Hour Per Week and BMI" and displays the distribution of individuals' Body Mass Index (BMI) about their weekly exercise hours. Analyzing the plot reveals no evident correlation between these two variables. The data points are widely spread out across the graph with a clear pattern, indicating that BMI values do not systematically increase or decrease with changes in exercise hours. The BMI values span approximately from 20 to 40, while weekly exercise hours range from 0 to 20, showing considerable variability in the data. This absence of a discernible trend suggests little to no correlation between exercise hours per week and BMI. If a correlation were present, one would expect the points to form a specific pattern, such as a straight line or a curve. Instead, the random distribution of points indicates that the relationship between exercise hours and BMI is neither strongly linear nor non-linear.

- Calculating the sample correlation coefficient by using Pearson product-moment correlation coefficient

The Pearson formula is:

$$r = \frac{\sum xy - (\sum x \sum y)/n}{\sqrt{[(\sum x^2) - (\sum x)^2/n][(\sum y^2) - (\sum y)^2/n]}}$$

r = Sample correlation coefficient, **n** = Sample size

x = Value of the independent variable, **y** = Value of the dependent variable

```
> cat("r =", r, "\n")
r = -0.02656121
```

Using RStudio, the calculated r-value is -0.02656121 which shows that the data points do not exhibit a clear negative (or positive) linear trend, suggesting that it might be close to 0, indicating a weak or no linear relationship between exercise hours per week and BMI.

- Test for correlation

Hypothesis:

$H_0 : \rho = 0$ (no linear correlation)

$H_1 : \rho \neq 0$ (linear correlation exists)

Calculate the test statistic:

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

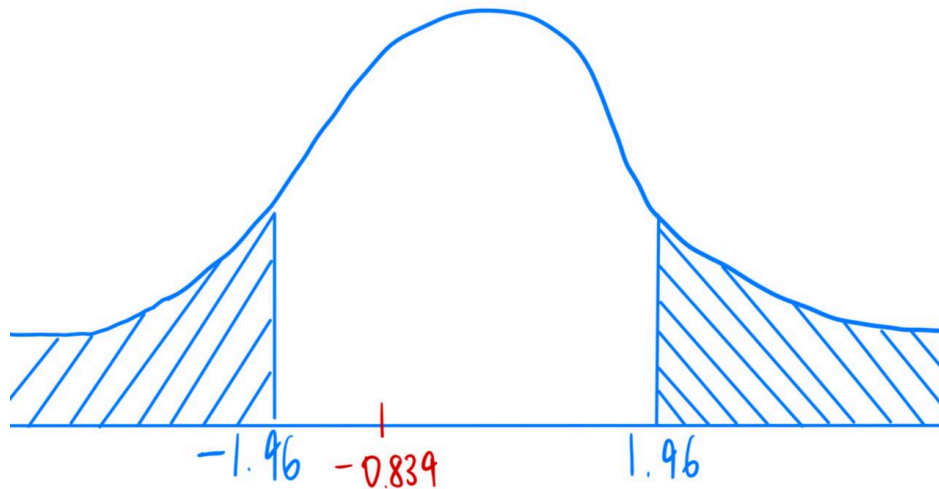
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	29.11070	0.39168	74.324	<2e-16
Exercise_Hour_Per_Week	-0.02878	0.03430	-0.839	0.402

Using RStudio, the calculated value of the test statistic is -0.839.

To find the critical value for $n=999$ and $df=997$ at the $\alpha=0.05$ significance level for a two-tailed test, we use a t-distribution table. Given the large degrees of freedom, we approximate using the standard normal distribution. The critical t-value for $\alpha/2=0.025$ is about 1.96 ($t_{0.025,997} \approx 1.96$).

The calculated t-value is -0.839, which is less than 1.96, we do not reject the null hypothesis. This is consistent with the P-value of 0.402, indicating no statistically significant relationship between exercise hours per week and BMI.



- Conclusion

Since the test statistic, $t = -0.839$ falls within the accepted region (between -1.96 and 1.96). Therefore, we fail to reject the null hypothesis because there is no sufficient evidence to conclude that there is a linear correlation.

This matches the P-value result of 0.402, indicating that the relationship between exercise hours per week and BMI is not statistically significant

```
> # Print the results
> cat("n =", n, "\n")
n = 999
> cat("df =", df, "\n")
df = 997
> cat("r =", r, "\n")
r = -0.02656121
> cat("P-value:", pVal, "\n")
P-value: 0.2008424
```

3.3 Regression Test

In the regression test, we measure the relationship between the cholesterol and heart rate for each individual with a sample size, $n = 899$. The independent variable, x is the cholesterol and the dependent variable, y is heart rate. The sample regression line provides an estimate of the population regression line.

Estimated Regression Model, $\hat{y}_i = b_0 + b_1x$

where,

$\hat{y}_i$ = estimated dependent variable value

b_0 = estimated of the regression intercept

b_1 = estimated of the regression slope

x = independent variable

Test statistic formula, $t = \frac{b_1 - \beta_1}{s_{b_1}}$; degree of freedom, $df = n - 2$

where,

b_1 = Sample regression slope coefficient

β_1 = Hypothesized slope

s_{b_1} = Estimator of the standard error of the slope

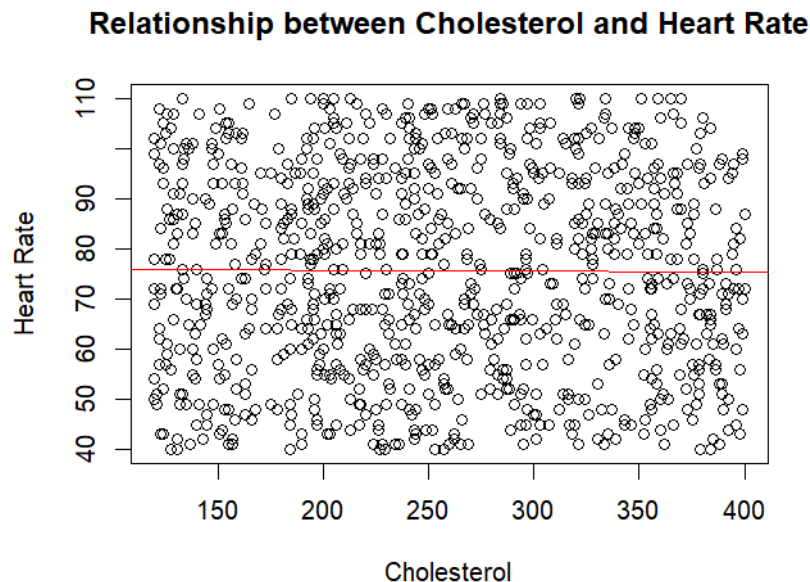


Figure 3.2: Scatter plot relationship between Cholesterol and Heart rate

From the summary, we can get the formula for estimated regression model which is

$$Y = 259.10 - 0.03197X$$

From the formula, the value of $b_0 = 259.10$ is the estimated value of y when the value of x is equal to 0. While the value of $b_1 = -0.03197$ measures the estimated change in the average value of heart rate, y as a result of one-unit change in cholesterol, x .

The coefficient of determination is the portion of the total variation in the dependent variable that is explained by variation in the independent variable.

$$R^2 = \frac{SSR}{SST} = \frac{\text{sum of squares explained by regression}}{\text{total sum of squares}}$$

From the RStudio we obtain the value of coefficient of determination, $R^2 = 6.446 \times 10^{-5}$. Since $0 < R^2 < 1$, the result shows a weaker linear relationship between the cholesterol, and heart rate, y . Some but not all of the variation in heart rate, y is explained by the variation in cholesterol, x .

Hypothesis statement:

Null hypothesis, $H_0: \beta_1 = 0$ (no linear relationship)

Alternative hypothesis, $H_1: \beta_1 \neq 0$ (linear relationship does exist)

where β_1 represent the population slope coefficient (hypothesized slope).

Test statistic :

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there exists a linear relationship between cholesterol and heart rate for each individual. From the RStudio, we obtain the sample regression slope coefficient, $b_1 = -0.002016$, the estimator of the standard error of the slope, $s_{b1} = 0.008386$, the test statistic value, $t = -0.24$, and the degree of freedom is $df = 897$. The probability value that we obtain is $P\text{-value} = 0.81$.

Conclusion:

Since $P\text{-value} = 0.81 > \alpha = 0.05$, we fail to reject the null hypothesis, H_0 . There is insufficient evidence to support the claim that there exists a linear relationship between cholesterol and heart rate for each individual. We can see that the residual and residual standard error, which is the difference between y and predicted y , is larger, and the R^2 is low, indicating no strong linear relationship. However, we cannot conclude that there is no relationship between heart rate and cholesterol, as this test specifically addresses the existence of a linear relationship.

3.4 ANOVA TEST

The purpose of the ANOVA test we conducted is to test for significant differences between means of cholesterol levels for two different genders. We use a 0.05 significance level to test the null hypothesis that cholesterol levels for two different genders will have the same mean. Due to the different sample size of each population being compared, we apply One-way ANOVA with Unequal Sample Sizes technique.

Test statistic formula:

- **Calculate Group Means and Overall Mean:**

Group means: $\bar{X}_i$ for each group i

Overall mean: $\bar{X} = \frac{1}{N} \sum_{i=1}^k \sum_{j=1}^{n_i} ij$

- **Sum of Squares Between Groups (SSB):**

$$SSB = \sum_{i=1}^k n_i (\bar{X}_i - \bar{X})^2$$

Where n_i is the number of observations in group i , k is the number of groups.

- **Sum of Squares Within Groups (SSW):**

$$SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2$$

- **Degrees of Freedom:**

Between groups: $df_{between} = k - 1$

Within groups: $df_{within} = N - k$, where N is the total number of observations.

- **Mean Squares:**

Mean Square Between Groups (MSB): $MSB = \frac{SSB}{df_{between}}$

Mean Square Within Groups (MSW): $MSW = \frac{SSW}{df_{within}}$

- **F-statistic:**

$$F = \frac{MSB}{MSW}$$

- **P-value:**

The p-value is calculated using the cumulative distribution function (CDF) of the F-distribution:

$$p\text{-value} = 1 - F_{CDF}(F, df_{between}, df_{within})$$

where F_{CDF} is the cumulative distribution function of the F-distribution with $df_{between}$ and df_{within} degrees of freedom.

Hypothesis statement:

Null hypothesis, $H_0: \mu_1 = \mu_2$ (same mean in both cholesterol levels)

Alternative hypothesis, $H_1: \mu_1 \neq \mu_2$ (different mean in cholesterol levels)

where

μ_1 represents the mean of cholesterol levels in females,

μ_2 represents the mean of cholesterol levels in males

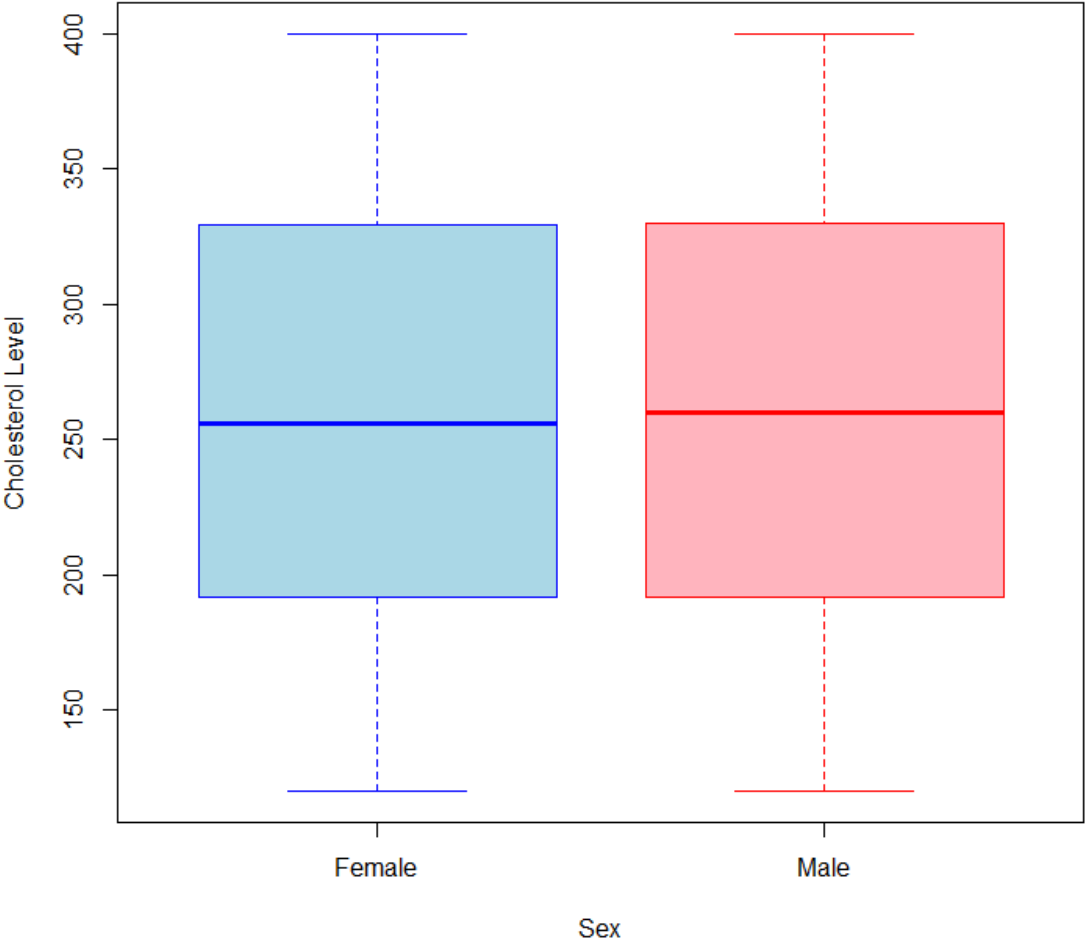
Test statistic:

A significance level of 0.05 ($\alpha = 0.05$) is used to test the claim that the means of cholesterol levels for two different genders will have the same mean. From the RStudio we obtain the test statistic value, $F = 0.508$. The probability value that we obtain is $P\text{-value} = 0.476$.

Conclusion:

Since $P\text{-value} > \alpha$ ($0.476 > 0.050$), thus fail to reject null hypothesis, H_0 . There is sufficient evidence to support the claims that the means of cholesterol levels for two different genders will have the same mean. This means that there is no significant evidence to suggest a difference in mean cholesterol levels between males and females in the dataset.

Cholesterol Levels by Sex



4.0 CONCLUSION

In conclusion, this project has been a new experience for all of us and valuable knowledge has been gained from it. It started with finding the datasets where we learned on how to choose suitable datasets for this project. Initially, we chose an unsuitable dataset which resulted in difficulty of making an analysis from it. From there, I agreed on finding a dataset that is suitable for our project. We made sure to learn from past mistakes and chose one that is suitable for this project. After that, in the pre-processing phase, we researched the variables provided from the dataset chosen to make sure that the variables to study on have relationships with one another to give meaningful insights on the data. Once we are done with the processing phase using R programming, we try to analyze the result from our calculations and make conclusions from it. There are times where the result is not what we were expecting and from there we reflect on why it is not what was expected by giving logical reasons to make a reasonable conclusion from our analysis.

5.0 Appendix

Raw dataset :

<https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset>

Processed dataset :

https://docs.google.com/spreadsheets/d/1Ox57l9E7dj-1UPTvE7QSiZ5Bc_gxL5V7brtwDZPLKi4/edit?usp=sharing

Presentation video :  **PSDA PROJECT 2**

Reflection post:

https://www.linkedin.com/posts/afif-shaqir-8b7a92285_reflection-activity-7214259466752466944-CAJZ?utm_source=share&utm_medium=member_desktop